



SAFETY DATA SHEET

(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 2015/830

Revision Date 29-Oct-2021

WAI2 - EGHS - EUROPEAN

Revision Number 4

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name 10ppm Fluoride with TISAB II Standard
Product No 040908
Pure substance/mixture Mixture

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent
Uses advised against No information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, importer, supplier

E-mail address info.water@thermo.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture**

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

2.2. Label elements**Precautionary Statements**

P202 - Do not handle until all safety precautions have been read and understood

2.3. Other hazards

No information available

General Hazards

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**3.1. Substances**

Component	EC No	CAS No	Weight percent	GHS Classification	REACH Reg. No
Water	EEC No. 231-791-2	7732-18-5	90 - 100%	Not classified	No information available
Sodium acetate	EEC No. 204-823-8	127-09-3	0 - 10%	-	No information available
Sodium chloride	EEC No. 231-598-3	7647-14-5	0 - 10%	-	No information available
trans-1,2-Diaminocyclohexane-N,N,N,N-tetraacetic acid monohydrate	-	125572-95-4	0 - 10%	-	No information available
Acetic acid	EEC No. 200-580-7	64-19-7	0 - 10%	Flam. Liq. 3 (H226) Skin Corr. 1A (H314)	No information available
Sodium fluoride	EEC No. 231-667-8	7681-49-4	0 - 10%	Acute Tox. 3 (H301) Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) (EUH032)	No information available
C.I. Acid Blue 9, disodium salt	EEC No. 223-339-8	3844-45-9	0 - 10%	-	No information available

Note *The exact percentage (concentration) of composition has been withheld as a trade secret

Full text of H- and EUH-phrases: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Use first aid treatment according to the nature of the injury. For further assistance, contact your local Poison Control Centre. Show this safety data sheet to the doctor in attendance.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.
Inhalation	Remove to fresh air. Get medical attention immediately if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.
Self-Protection of the First Aider	No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	None reasonably foreseeable
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4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable Extinguishing Media

No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapours.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment as required.
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6.2. Environmental precautions

Environmental Precautions	Should not be released into the environment. See Section 12 for additional Ecological Information.
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6.3. Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up Soak up with inert absorbent material. Pick up and transfer to properly labelled containers.

Reference to other sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Advice on safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Storage Conditions

Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Protect from direct sunlight.

7.3. Specific end use(s)

Specific Use(s)

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure Limits

Component	European Union	The United Kingdom	France	Spain	Germany
Acetic acid 64-19-7	TWA: 25 mg/m ³ (15min) TWA: 10 ppm (15min) STEL: 50 mg/m ³ (8h) STEL: 20 ppm (8h)	STEL: 20 ppm 15 min STEL: 50 mg/m ³ 15 min TWA: 10 ppm 8 hr TWA: 25 mg/m ³ 8 hr	TWA / VME: 10 ppm (8 heures). TWA / VME: 25 mg/m ³ (8 heures). STEL / VLCT: 20 ppm. STEL / VLCT: 50 mg/m ³ .	STEL / VLA-EC: 20 ppm (15 minutos). STEL / VLA-EC: 50 mg/m ³ (15 minutos). TWA / VLA-ED: 10 ppm (8 horas) TWA / VLA-ED: 25 mg/m ³ (8 horas)	TWA: 10 ppm (8 Stunden). AGW - exposure factor 2 TWA: 25 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 10 ppm (8 Stunden). MAK TWA: 25 mg/m ³ (8 Stunden). MAK Höhepunkt: 20 ppm Höhepunkt: 50 mg/m ³
Sodium fluoride 7681-49-4	-	STEL: 7.5 mg/m ³ 15 min TWA: 2.5 mg/m ³ 8 hr	TWA / VME: 2 mg/m ³ (8 heures). TWA / VME: 2.5 mg/m ³ (8 heures). indicative limit	TWA / VLA-ED: 2.5 mg/m ³ (8 horas)	TWA: 1 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 1 mg/m ³ (8 Stunden). MAK Haut
Component	Italy	Portugal	The Netherlands	Finland	Denmark
Acetic acid 64-19-7	TWA: 25 ppm 8 ore. Time Weighted	STEL: 20 ppm 15 minutos	STEL: 50 mg/m ³ 15 minuten	TWA: 5 ppm 8 tunteina TWA: 13 mg/m ³ 8	TWA: 10 ppm 8 timer TWA: 25 mg/m ³ 8

	Average TWA: 10 mg/m ³ 8 ore. Time Weighted Average STEL: 50 mg/m ³ 15 minuti. Short-term STEL: 20 ppm 15 minuti. Short-term	STEL: 50 mg/m ³ 15 minutos TWA: 10 ppm 8 horas TWA: 25 mg/m ³ 8 horas	TWA: 25 mg/m ³ 8 uren	tunteina STEL: 10 ppm 15 minuutteina STEL: 25 mg/m ³ 15 minuutteina	timer
Sodium fluoride 7681-49-4	-	TWA: 2.5 mg/m ³ 8 horas	-	-	-
Component	Austria	Switzerland	Poland	Norway	Ireland
Acetic acid 64-19-7	MAK-KZGW: 20 ppm 15 Minuten MAK-KZGW: 50 mg/m ³ 15 Minuten MAK-TMW: 10 ppm 8 Stunden MAK-TMW: 25 mg/m ³ 8 Stunden	STEL: 20 ppm 15 Minuten STEL: 50 mg/m ³ 15 Minuten TWA: 10 ppm 8 Stunden TWA: 25 mg/m ³ 8 Stunden	STEL: 50 mg/m ³ 15 minutach TWA: 25 mg/m ³ 8 godzinach	TWA: 10 ppm 8 timer TWA: 25 mg/m ³ 8 timer STEL: 20 ppm 15 minutter. value from the regulation STEL: 50 mg/m ³ 15 minutter. value from the regulation	TWA: 20 ppm 8 hr. TWA: 50 mg/m ³ 8 hr. STEL: 20 ppm 15 min STEL: 50 mg/m ³ 15 min
Sodium fluoride 7681-49-4	-	-	-	TWA: 0.5 mg/m ³ 8 timer	-

Derived No Effect Level (DNEL) See table for values

Predicted No Effect Concentration (PNEC) See values below

8.2. Exposure controls

Engineering Measures None under normal use conditions

Personal protective equipment

Eye/face Protection Wear safety glasses with side shields (or goggles).

Skin and body protection Wear protective gloves/protective clothing.

Respiratory Protection No protective equipment is needed under normal use conditions.
Recommended Filter type: Particle filter.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Liquid
Appearance	Blue
Odour	vinegar-like
Odour Threshold	No information available
pH	3.8
PH Range	2.3 - 5.3

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling point/range	~ 100 °C / 212 °F	
Flash Point	No information available	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit	No information available	
Vapour pressure	No information available	

Vapour Density	No information available
Specific Gravity	No information available
Water Solubility	Soluble in water
Solubility in other solvents	No information available
Partition coefficient	No information available
Autoignition Temperature	-
Decomposition Temperature	No information available
Kinematic viscosity	No information available
Dynamic viscosity	No information available
Explosive Properties	No information available
Oxidising Properties	No information available

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	0.8
Density	No information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY**10.1. Reactivity**

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact	None
Sensitivity to Static Discharge	None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapours

SECTION 11: TOXICOLOGICAL INFORMATION**11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008****Acute Toxicity****Product Information**

Inhalation	No information available
Eye Contact	No information available
Skin Contact	No information available
Ingestion	No information available

Unknown Acute Toxicity 0 % of the mixture consists of ingredient(s) of unknown toxicity.**The following values are calculated based on chapter 3.1 of the GHS document**
ATEmix (oral) 24,653.00 mg/kg

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	LD50 > 90 mL/kg (Rat)		
Sodium acetate	LD50 = 3530 mg/kg (Rat)	LD50 > 10 g/kg (Rabbit)	LC50 > 30 g/m ³ (Rat) 1 h
Sodium chloride	LD50 = 3 g/kg (Rat)	LD50 > 10000 mg/kg (Rabbit)	LC50 > 42 mg/L (Rat) 1 h
Acetic acid	LD50 = 3310 mg/kg (Rat)	LD50 = 1060 mg/kg (Rabbit)	LC50 = 11.4 mg/L (Rat) 4 h
Sodium fluoride	LD50 = 52 mg/kg (Rat)	LD50 = 175 mg/kg (Rat)	
C.I. Acid Blue 9, disodium salt	LD50 > 2 g/kg (Rat)		

Skin Corrosion/Irritation	No information available
Serious eye damage/eye irritation	No information available
Sensitisation	No information available
Mutagenic Effects	No information available
Carcinogenic effects	No information available
Reproductive Effects	No information available
STOT - single exposure	No information available
STOT - repeated exposure	No information available
Aspiration hazard	No information available

SECTION 12. ECOLOGICAL INFORMATION

12.1. Toxicity

1% of the mixture consists of components(s) of unknown hazards to the aquatic environment

Component	Freshwater Algae	Freshwater Fish	Water Flea
Sodium acetate	-	LC50: > 100 mg/L, 96h semi-static (Danio rerio)	EC50: > 1000 mg/L, 48h (Daphnia magna)
Sodium chloride	-	LC50: 6420 - 6700 mg/L, 96h static (Pimephales promelas) LC50: 4747 - 7824 mg/L, 96h flow-through (Oncorhynchus mykiss) LC50: 6020 - 7070 mg/L, 96h static (Pimephales promelas) LC50: = 12946 mg/L, 96h static (Lepomis macrochirus) LC50: 5560 - 6080 mg/L, 96h flow-through (Lepomis macrochirus) LC50: = 7050 mg/L, 96h semi-static (Pimephales promelas)	EC50: 340.7 - 469.2 mg/L, 48h Static (Daphnia magna) EC50: = 1000 mg/L, 48h (Daphnia magna)
Acetic acid	-	LC50: = 75 mg/L, 96h static (Lepomis macrochirus) LC50: = 79 mg/L, 96h static (Pimephales promelas)	EC50: = 65 mg/L, 48h Static (Daphnia magna)
Sodium fluoride	EC50: = 850 mg/L, 72h static (Desmodesmus subspicatus) EC50: = 272 mg/L, 96h	LC50: 38 - 68 mg/L, 96h static (Oncorhynchus mykiss) LC50: = 180 mg/L, 96h semi-static	EC50: = 98 mg/L, 48h Static (Daphnia magna) EC50: = 338 mg/L, 48h (Daphnia)

	(Pseudokirchneriella subcapitata)	(Pimephales promelas) LC50: = 830 mg/L, 96h semi-static (Lepomis macrochirus) LC50: > 530 mg/L, 96h (Lepomis macrochirus)	magna)
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12.2. Persistence and degradability

No information available

12.3. Bioaccumulative potential

No information available

12.4. Mobility in soil

No information available

Mobility

Component	log Pow
Acetic acid 64-19-7 (0 - 10%)	-0.17
C.I. Acid Blue 9, disodium salt 3844-45-9 (0 - 10%)	<-6.4

12.5. Results of PBT and vPvB assessment

No information available

12.6. Endocrine disrupting properties

No information available

Endocrine Disruptor Information

No information available

SECTION 13. DISPOSAL CONSIDERATIONS**13.1. Waste treatment methods****Waste from Residues/Unused Products**

Disposal should be in accordance with applicable regional, national and local laws and regulations.

Contaminated Packaging

Improper disposal or reuse of this container may be dangerous and illegal.

SECTION 14: TRANSPORT INFORMATION**IMDG/IMO**

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Marine Pollutant	Not Applicable
14.6 Special Provisions	None
14.7 Transport in bulk according to Annex II of MARPOL and the IBC Code	No information available

ICAO

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Environmental hazard	Not Applicable

14.6 Special Provisions None

IATA

14.1 UN-No	Not Regulated
14.2 Proper Shipping Name	Not Regulated
14.3 Hazard Class	Not Regulated
14.4 Packing Group	Not Regulated
14.5 Environmental hazard	Not Applicable
14.6 Special Provisions	None

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

WGK Classification Water endangering class = 1 (self classification)

European Union

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

International Inventories

United States of America Inventory	Complies
CANINV	Complies
EINECS/ELINCS	Does not Comply
ENCS	Does not Comply
IECSC	Does not Comply
KECL	Does not Comply
PICCS	Does not Comply
AICS	Does not Comply

Legend:

USINV/ TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
CANINV/ DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
ENCS - Japanese Existing and New Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
AICS - Australian Inventory of Chemical Substances

15.2. Chemical safety assessment

A chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION

Key or legend to abbreviations and acronyms used in the safety data sheet

Full text of H-Statements referred to under section 3

H301 - Toxic if swallowed
H315 - Causes skin irritation
H319 - Causes serious eye irritation
EUH032 - Contact with acids liberates very toxic gas

Legend - SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

TWA	TWA (time-weighted average)	STEL	STEL (Short Term Exposure Limit)
Ceiling	Maximum limit value	*	Skin designation

Prepared By Regulatory Affairs

Prepared For Thermo Fisher Scientific Inc.

Issue Date No information available

Revision Date 29-Oct-2021

Reason for revision SDS sections updated.

Training Advice Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

Disclaimer

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet